

Distributed Event-Triggered Estimation Over Sensor Networks: A Survey

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Abstract—An event-triggered mechanism is of great efficiency in reducing unnecessary sensor samplings/transmissions and, thus, resource consumption such as sensor power and network bandwidth, which makes distributed event-triggered estimation a promising resource-aware solution for sensor network-based monitoring systems. This paper provides a survey of recent advances in distributed event-triggered estimation for dynamical systems operating over resource-constrained sensor networks. Local estimates of an unavailable state signal are calculated in a distributed and collaborative fashion based on only invoked sensor data. First, several fundamental issues associated with the design of distributed estimators are discussed in detail, such as estimator structures, communication constraints, and design methods. Second, an emphasis is laid on recent developments of distributed event-triggered estimation that has received considerable attention in the past few years. Then, the principle of an event-triggered mechanism is outlined and recent results in this subject are sorted out in accordance with different event-triggering conditions. Third, applications of distributed event-triggered estimation in practical sensor network-based monitoring systems including distributed grid-connected generation systems and target tracking systems are provided. Finally, several challenging issues worthy of further research are envisioned.

Index Terms—Attack detection, communication constraint, distributed estimation, event-triggered mechanism, islanding detection, secure estimation, sensor networks, set-membership filtering, target tracking.

I. INTRODUCTION

SENSOR networks have been recognized as a promising technology that is employed for data collection, state/parameter estimation, target tracking, and remote system

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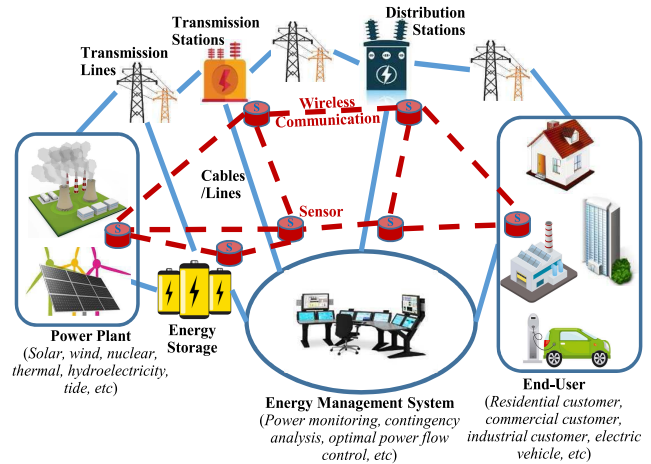


Fig. 1. Smart grid monitored by a sensor network.

monitoring purposes because of their ease of deployment, cost-effectiveness, flexibility, reliability, and accuracy [1]. A sensor network consists of a large number of smart sensor nodes, which are dispersedly deployed in a monitoring field (also known as region of interest) to cooperatively observe physical or environmental conditions. Sensor nodes often equip radio transceivers allowing them to communicate with each other via wireless channels and are fitted with onboard processors, enabling them to carry out computation of the gathered data.

Sensor networks have revolutionized sensing and communication in many practical applications. For example, recent advancements in enhanced sensing, communication, and computing technologies have rendered the application of sensor networks in smart grids feasible, as illustrated in Fig. 1. Different components of the grid are linked together with smart sensors and wireless communication paths in the sense that interoperability can be guaranteed between generation, storage, distribution, transmission, and end users. In today's power grids, major concerns lie in the reliable and time-critical information exchange, real-time monitoring, and reaction. In this respect, intelligent and low-cost monitoring enabled by real-time sensing and communication technologies has become essential to maintain safety, reliability, efficiency, and uptime of the envisioned smart grids [2]–[4]. Clearly, sensor networks play a vital role in addressing the issues of reliable monitoring and uninterrupted protection for smart grids through collaborative and distributed information processing. Other practical

examples of sensor network applications can be found in a wide range of fields, such as *military* (battlefield surveillance, enemy locating, and tracking), *industry* (factory and process automation monitoring), *civilian* (building monitoring and home automation), *environment* (wildlife habitat monitoring, fire detection in a forest, and toxic chemical detection in a contaminated field), *traffic* (traffic conditions detection), and *health* (personal and wellness monitoring). However, it should be stressed that the realization of these currently designed and envisioned applications relies on efficient and reliable communication capabilities of the deployed sensor networks [2].

To implement the sensor network technology in practical applications, there are several challenges, ranging from design cost, hardware constraints, and environment to transmission media, communication protocols, resource and design constraints, and intelligent information processing, that should be well addressed [5]–[8]. Among these challenges, there are two critical issues worthy of special consideration.

The first issue is *how to deal with the resource and design constraints of sensor networks*. Sensor networks pose various resource and design constraints because of their unique features and application requirements including, but not limited to, the following.

- 1) Limited amount of battery, short sensing/communication range, restricted processing and computational capabilities, and small memory.
- 2) Communication constraints, such as data transmission/communication delays, data losses, time-varying sensing/network topology and so on.
- 3) Security goals in terms of data confidentiality, integrity, availability, authorization, authentication, nonrepudiation, and freshness [9].
- 4) Network self-organization: Given the large number of nodes and their potential deployment in a hostile environment, node failures may occur (either from a lack of power and bandwidth or from physical destruction) and new nodes may enter the network. The network thus must be able to self-organize in order to continue to function.
- 5) Data aggregation/fusion: To avoid excessive energy consumption and network congestion, sensors should efficiently use the limited resources to perform a certain level of aggregation/fusion on their raw data and broadcast data to only neighboring sensors.
- 6) Collaboration and coordination: Limited by its sensing/communication capabilities, each sensor can only observe partial information of interest and exchange data only with its neighbors. Sensors should collaborate among themselves and coordinate their local observations to enhance the global observability of the sensor network.

The second issue is *how to achieve effective and efficient distributed information processing of data gathered from scattered sensors*. In most sensor network applications, sensors can provide users access to observed data anytime and anywhere by collecting and disseminating data. Whereas users are more interested in processed data carrying useful information of

interest rather than raw data because raw data may be contaminated by some unwanted signals. Besides, to meet a finite energy budget, raw sensor data should be processed as much as possible inside the network in order to reduce the number of bits transmitted, particularly over longer distances [10]. Lying in the core part of distributed information processing is the *distributed estimation* problem that aims at computing local estimates of an inaccessible full state signal of a dynamical system or an unknown parameter of a physical phenomenon from raw sensor observations/measurements, sometimes also referred to as distributed parameter estimation [11], [12]. Both terms are interchangeably used in the literature, while they have different problem statements and, thus, require distinct analysis and design techniques. We prefer the term “distributed estimation” because this paper focuses on the state-space description of a dynamical system.

The literature on distributed estimation over sensor networks is quite versatile and considers many different system models, problem formulations, and resource and design constraints. For example, in [13] and [14], the problems of multi-input–multi-output decision fusion and distributed detection in wireless-sensor networks over fading channels are considered. In [15], various distributed sequential Bayesian state estimation approaches and algorithms in multiagent networks are surveyed, classified, and compared, where the focus is on decentralized agent networks without a central processing unit. In [16], a bibliographical review on distributed Kalman filtering is provided, where a central Kalman filter is involved. Different approaches involved in distributed Kalman filtering are classified and discussed. In [17], recent advances on distributed state estimation for stochastic systems over sensor networks are presented, in which an emphasis is laid on various performance requirements and randomly occurring network-induced phenomena, such as sensor saturation, signal quantization, missing measurements, randomly occurring nonlinearities, stochastic communication delays, and packet dropouts. However, there is a lack of comprehensive reviews and summaries of distributed resource-aware estimation in the context of event-triggered sampling/transmission. The aim of this paper is to provide a survey of recent advances in distributed event-triggered estimation for dynamical systems operating over resource-constrained sensor networks. The reminder of this paper is organized as follows. Section II outlines a general framework of distributed estimation and details some fundamental issues for designing suitable distributed estimators. Section III presents the recent developments and research results in distributed event-triggered estimation. Section IV presents two application examples on how distributed event-triggered estimation can be employed in practical sensor network-based monitoring systems. Section V concludes this paper with discussions on some challenging issues worthy of future research.

II. FUNDAMENTAL ISSUES OF DISTRIBUTED ESTIMATION

As one of the key issues in the study of dynamical systems, estimation allows a system to acquire full knowledge of its current state, as accurate as possible and, thus, make informed

control decisions [18]. However, estimation is challenging because the complete state of a dynamical system is not always measurable and accessible in many practical scenarios, for example, the system model has uncertainties or is contaminated by unknown external disturbance and noise, or the network transmission channels are corrupted by unknown communication noise, or it is difficult and costly to measure the full information of the system's state due to limited sensing capability and high-order system dynamics [19].

For simplicity of demonstration, we consider a dynamical system of the following linear state-space form:

$$s^\Delta(t) = As(t) + Bw(t), \quad s(0) = s_0 \quad (1)$$

where $s(t) \in \mathbb{R}^{n_s}$ denotes the system's state; $w(t) \in \mathbb{R}^{n_w}$ represents the process noise (or external disturbance input); $s^\Delta(t)$ denotes the differential operator $\dot{s}(t)$ in continuous-time case and the one step ahead operator $s(t+1)$ in discrete-time case, respectively; $z(t) = Es(t) \in \mathbb{R}^{n_z}$ stands for the target state to be estimated; s_0 is the initial state; and A , B , and E are generally time-varying matrices of appropriate dimensions.

It is assumed that system (1) is monitored by a group of N spatially dispersed sensors. These N sensors form a cooperative sensing/communication network whose interaction topology is modeled by a digraph $\mathcal{G} = (\nu, \mathcal{E}, \mathcal{A})$, where $\nu = \{1, 2, \dots, N\}$ is an index set of N sensors, $\mathcal{E} = \{(i, j) \mid i, j \in \nu\}$ is an edge set of interaction links, and $\mathcal{A} = [a_{ij}]_{N \times N}$ is a weighted adjacency matrix with weights $a_{ij} > 0$ if $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. It is common to include self-loops, which gives rise to $(i, i) \in \mathcal{E}$ or $a_{ii} \neq 0$. Any sensor residing within the sensing/communication range of sensor i is called a neighbor of sensor i . The neighbor set sensor i is thus denoted by $\mathcal{N}_i = \{j \in \nu \mid d_{ij} \leq r_i\}$, where d_{ij} is the distance between sensor i and sensor j ; and r_i is the diameter of the range of sensor i . A directed edge $(i, j) \in \mathcal{E}$ means that sensor i can receive information from the neighboring sensor j , but not vice versa. At any time t , the observation/measurement model of sensor i for system (1) is given by

$$y_i(t) = C_i s(t) + D_i v_i(t) \quad (2)$$

where $y_i(t) \in \mathbb{R}^{n_y}$ is the measurement output on sensor i ; $v_i(t) \in \mathbb{R}^{n_v}$ represents the sensor measurement/communication noise; and C_i and D_i are generally time-varying matrices of appropriate dimensions.

A general description of the *distributed estimation problem* is stated as follows. Consider system (1), given the local sensor measurement $y_i(t)$ of the form (2), $\forall i \in \mathcal{V}$, the objective is to design N distributed estimators of suitable forms such that each estimator can determine a local estimate $z_i(t)$ of the system's target state $z(t)$ at every instant of time t under the effects of disturbance and noise, while simultaneously achieving a prescribed performance index.

In the following, some fundamental issues that arise with the design of distributed estimators are elaborated.

A. Estimator Structures

Depending on whether there is a fusion center (FC), there are two prevalent structures of distributed estimators.

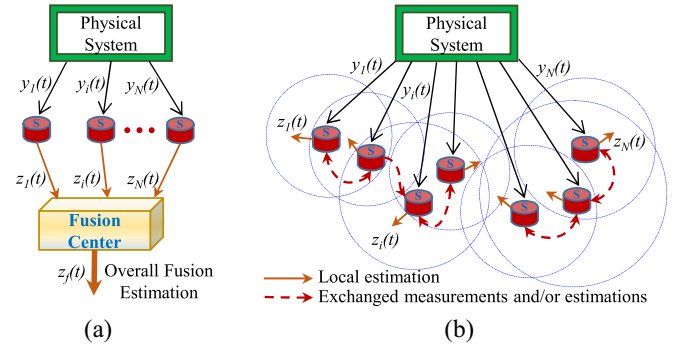


Fig. 2. Distributed estimation (a) with an FC and (b) without an FC.

- 1) When an FC is present, a common distributed estimator structure is of the following form [20]:

$$x_i^\Delta(t) = Ax_i(t) + K_i(\tilde{y}_i(t) - C_i x_i(t)) \quad (3)$$

where $x_i(t) \in \mathbb{R}^{n_s}$ denotes the state of estimator i ; $\tilde{y}_i(t) \in \mathbb{R}^{n_y}$ represents the actual measurement received from sensor i ; $z_i(t) = Ex_i(t) \in \mathbb{R}^{n_z}$ represents the local sensor estimation of $z(t)$ computed by estimator i ; $x_i^\Delta(t)$ denotes the differential operator $\dot{x}_i(t)$ in the continuous-time case and the one step ahead operator $x_i(t+1)$ in the discrete-time case, respectively; K_i is the estimator gain matrix to be designed; and $x_i(0) = x_{i,0}$ is the initial state of estimator i . In this case, there is no intercommunication among sensors; communication occurs only between sensors and the FC via some shared wireless communication medium; each sensor acts as an estimator capable of computing a local estimate. The FC then aggregates locally processed data by employing some suitable optimal fusion rules and finally generates an overall state estimate described by $z_f(t) = \sum_{i=1}^N W_i z_i(t)$, also see Fig. 2(a), where $z_f(t)$ denotes the overall fusion estimate and W_i is the optimal weighting matrix to be determined. Although the distributed estimation framework with an FC generally leads to an optimal global estimate, it has at least three drawbacks: 1) susceptible to a critical failure point at the FC; 2) high energy expenditure for information collecting and communication at the FC; and 3) a lack of collaboration and coordination among sensors.

- 2) When an FC is absent, the sensor network itself is responsible for processing the raw collected data and is considered to be an ad-hoc sensor network; sensors have no hierarchy in the network and, thus, have the same type of processing and communication [21]. Sensors act also as local estimators and communicate either their local observations or estimates or both of them with only their neighbors through some shared wireless communication medium, see Fig. 2(b). Such a wireless intersensor communication brings in a major challenge of the distributed estimation problem, that is, the integrated design of local information processing operations and intelligent intersensor communication strategies in order to strike a tradeoff among desirable system performance

and satisfactory communication performance such as energy efficiency and bandwidth reduction.

Denote by $\tilde{x}_j(t)$ (or $\tilde{y}_j(t)$) the actual transmitted estimate (or measurement) on sensor j , and L_{ij} , K_{ij} , L_i , and K_i the estimator gain matrices on sensor i (particularly, one may let $L_{ij} \equiv L_i$ or $K_{ij} \equiv K_i$ to get variants). Note that in most practical scenarios $\tilde{x}_j(t) \neq x_j(t)$ (or $\tilde{y}_j(t) \neq y_j(t)$), due to unreliable wireless communication channels, for example, in the presence of sampling, quantization, delay, loss, etc. The existing distributed estimator structures without an FC can be arguably classified into the following three types, characterized by aggregation strategies.

Type-I: Distributed estimators based on *aggregated measurements*. A typical distributed estimator structure of this type, indexed by $i \in \mathcal{V}$, has the following form:

$$\dot{x}_i^\Delta(t) = L_i x_i(t) + \sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} \tilde{y}_j(t) \quad [22]–[24] \quad (4)$$

where $\sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} \tilde{y}_j(t)$ represents the aggregated measurement term from all sensor i 's neighbors $j \in \mathcal{N}_i$.

Type-II: Distributed estimators based on *aggregated estimates*. A typical distributed estimator structure of this type, indexed by $i \in \mathcal{V}$, has the following forms:

$$\dot{x}_i^\Delta(t) = \sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} (\tilde{x}_i(t) - \tilde{x}_j(t)) + K_i \tilde{y}_i(t) \quad [25] \quad (5)$$

$$\begin{aligned} \dot{x}_i^\Delta(t) = & A x_i(t) + \sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} (\tilde{x}_i(t) - \tilde{x}_j(t)) \\ & + K_i (\tilde{y}_i(t) - C_i \tilde{x}_i(t)) \quad [26]–[32] \end{aligned} \quad (6)$$

where $\sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} (\tilde{x}_i(t) - \tilde{x}_j(t))$ represents the aggregated estimate term from all sensor i 's neighbors $j \in \mathcal{N}_i$.

Type-III: Distributed estimators based on both *aggregated measurements and estimates*. A typical distributed estimator structure of this type, indexed by $i \in \mathcal{V}$, has the following various forms:

$$\dot{x}_i^\Delta(t) = A x_i(t) + \sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} (\tilde{y}_j(t) - C_j \tilde{x}_j(t)) \quad [33]–[35] \quad (7)$$

$$\begin{aligned} \dot{x}_i^\Delta(t) = & A x_i(t) + \sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} (\tilde{x}_i(t) - \tilde{x}_j(t)) \\ & + \sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} (\tilde{y}_j(t) - C_j \tilde{x}_j(t)) \quad [36]–[38] \end{aligned} \quad (8)$$

$$\dot{x}_i^\Delta(t) = \sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} \tilde{x}_j(t) + \sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} \tilde{y}_j(t) \quad [39]–[42] \quad (9)$$

$$\begin{aligned} \dot{x}_i^\Delta(t) = & \sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} \tilde{x}_j(t) \\ & + \sum_{j \in \mathcal{N}_i} a_{ij} K_{ij} (\tilde{y}_j(t) - C_j \tilde{x}_j(t)) \quad [43], [44] \end{aligned} \quad (10)$$

$$\dot{x}_i^\Delta(t) = \sum_{j \in \mathcal{N}_i} a_{ij} (A \tilde{x}_j(t) + K_j (\tilde{y}_j(t) - C_j \tilde{x}_j(t))) \quad [45], [46]. \quad (11)$$

It should be noted that:

- 1) in contrast to the centralized estimation, the design of distributed estimation algorithms in a sensor network setting should be coupled with the underlying

communication topology $\mathcal{G}$. In other words, each sensor should be able to reflect such a coupling feature by incorporating individual data from itself but also aggregated data from its neighbors. Clearly, the estimators (4)–(11) can be divided into two parts: one denotes the measurement and/or estimate from sensor i itself and the other represents the communication between sensor i and all of its underlying neighbors;

- 2) whether sensor measurements or estimates should be shared among neighboring sensors is determined by practical design constraints and system performance requirements. Whereas, letting sensor estimates rather than measurements be exchanged generally provides certain freedom and flexibility to choose the information sent through the network [26];
- 3) the above aggregation strategies for either sensor measurements or sensor estimates share a common interaction topology characterized by $\mathcal{G}$. More generally, one may design a distributed estimator by considering a *heterogeneous interaction network* where the interaction topologies of measurements and estimates are different, which is reasonable in certain practical situations because of different sensor communication capacities and an unreliable wireless communication environment;
- 4) for estimators (5), (6), and (8), the term $\sum_{j \in \mathcal{N}_i} a_{ij} L_{ij} (\tilde{x}_i(t) - \tilde{x}_j(t))$ is introduced, which was first proposed by Olfati-Saber and Shamma [47] for evaluating the average consensus on sensor measurements. Since then, many results on distributed consensus estimation have been published, see the recent survey [48] and references therein. Such a distributed consensus estimation strategy is particularly useful for computing averages as the fundamental objective is that all sensors should guarantee the same estimate in steady state [49];
- 5) different from the estimators (4), (5), (9), and (10), the estimators (6)–(8) and (11), also known as Luenberger-type estimators, do *not* require that the system matrix A of the original system (1) be Hurwitz or Schur stable. Whereas these estimators need to have the same order of the system, that is, $s(t)$ and $x_i(t)$ both belong to $\mathbb{R}^{n_s}$, which leads to the full-order estimator case. However, such a restriction is *not* imposed on the estimators (4), (5), (9), and (10) so that reduced-order estimators can be designed.

B. Communication Constraints

The backbone of a sensor network lies in its wireless medium, which provides a two-way communication of sensor observations among the spatially distributed nodes. The introduction of the wireless communication medium offers several benefits for sensor networks, including flexibility, scalability, and manipulability that are beyond the capability of traditional communication networks. Nevertheless, the communication network can also introduce certain levels of constraints in terms of sampling, delay, data loss, quantization and so on [50]–[55], when sensor observations are disseminated via the wireless channels.

TABLE I
REFERENCES SIMULTANEOUSLY CONSIDERING MULTIPLE
COMMUNICATION CONSTRAINTS

&	(ii)	(iii)	(iv)	(v)	(vi)
(i)	[26], [68]		[68], [71]	[71]	
(ii)		[40], [69], [70]	[41]	[57], [69]	
(vii)					[72]

The communication constraints that have been extensively addressed in the existing distributed estimation literature can be roughly categorized as follows:

- i) transmission/communication delays;
- ii) data losses (or data packet dropouts, missing measurements, partial information);
- iii) quantization errors;
- iv) aperiodic sampling/transmission intervals;
- v) time-varying sensing/network topology;
- vi) fading network channels;
- vii) competition of multiple nodes accessing network.

It is well acknowledged that these communication constraints may deteriorate the estimation performance or even destabilize the estimation error system. Therefore, modeling of these constraints is of great significance so that one can quantitatively understand how they influence the estimation performance. Different modeling methods can lead to various analysis and design criteria with discrepant conservatism, which may yet be demanded in diverse practical scenarios. It is also worth noting that much of the available literature on distributed estimation considers only one of the above-mentioned types of constraints. For example, the effects of *delays* are investigated in [22] and [56]; of *data losses* in [32], [33], [42], [43], and [57]–[63]; of *quantization* in [38] and [64]; of *aperiodic sampling/transmission* in [34]; of *time-varying topology* in [29]; and of *node scheduling or selection* in [30] and [65]–[67], respectively. In contrast, few results are reported to deal with combinations of these communication constraints in the context of distributed estimation. References that study simultaneously two or more types of the constraints aforementioned are categorized in Table I. Since, for a practical sensor network, the different types of constraints listed above may occur simultaneously, there is a necessity for distributed estimation algorithms accommodating multiple constraints. Until now, several systematic reviews on communication constraints have unfolded with respect to the modeling and the different approaches for handling them. We refer readers to the surveys [73] and [74] on networked control systems and [75] on distributed networked control systems.

C. Design Methods

Some noise may enter practical dynamical systems in different ways and various forms, appearing in both the system state equation and the sensor measurement equation. With noise acting as driving force in the equations, the system dynamics can be drastically modified and even destabilized [76]. It is thus essential to filter out the unwanted noise signal from the available sensor measurements and minimize the influences so that

the best estimate can be determined. However, before embarking on an exposition of the estimator design, one may wonder about the type of noise. Generally, the noise that is encountered in practical dynamical systems is of two major types: 1) *additive noise*, which could be deemed as an external disturbance and is simply added to the basic deterministic part of the state equation of the system and/or sensor measurement equations, for example, $w(t)$ and $v_i(t)$ in (1) and (2), respectively and 2) *multiplicative (or state-dependent) noise*, which is regarded as an internal dynamic disturbance coupled with the system's state mainly due to unmodeled dynamics. For example, in the discrete-time case, the following two multiplicative terms $F_1 s(t)\theta(t)$ and $F_2 s(t)\vartheta_i(t)$ can be added to (1) and (2), respectively. Clearly, these terms are dependent on the value of the state variable $s(t)$ through the multiplicative noise $\theta(t)$ and $\vartheta_i(t)$. It will be more difficult to deal with multiplicative noise than additive noise because the former usually allows its statistics description to be unknown but depend on the real state of the system.

In general, modeling noise is challenging because one does not have access to noise information itself. Rather, one can only acquire locally available sensor observations which are perturbed by one or more types of noise. To conduct the quantitative performance analysis of a state estimation algorithm, it is thus required that noise should have some well-defined properties, such as known statistical characteristics and bounded energy. In what follows, some commonly made assumptions on noise are summarized.

- 1) *Random Noise With Known Statistical Properties*: For example, multiplicative noise variables $\theta(t)$ and $\vartheta_i(t)$ are often assumed to be zero-mean Gaussian white noise with known covariances.
- 2) *Unknown Nonrandom Noise But With Bounded Energy (also known as Energy-Bounded Noise)*: The most commonly encountered notion of signal energy defined on $\mathbb{R}$ is the l_2 - or L_2 -norm. For example, in the discrete-time case, the l_2 -norm of the additive noise variable $w(t)$ in (1) is defined as $\|w(t)\|_2 = (\sum_{t=0}^{\infty} w^T(t)w(t))^{1/2}$.
- 3) *Unknown Nonrandom Noise But With Bounded Magnitude (also known as UBB Noise)*: With no assumption on the noise statistics, UBB noise only requires the least amount of *a priori* knowledge of the noise, that is, a known bound. For example, in the discrete-time case, the additive noise variable $w(t)$ in (1) is assumed to be confined to an ellipsoid $W_t \triangleq \{w(t) : w^T(t)R_t^{-1}w(t) \leq 1\}$, where R_t is a positive-definite time-varying matrix.

In order to measure the response quality of the estimation error dynamics, several performance indices have been explored in the literature to evaluate the estimation error $e_i(t) = z_i(t) - z(t)$. It should be noted that how to formalize a suitable performance index and a notion of optimality depends on the estimators selected, the noise type and assumption, and the actual system model under consideration. In the sequel, we highlight the three major classes of performance indices for discrete-time system models, whereas the counterparts of continuous-time systems are a trivial extension.

TABLE II

TYPICAL DISTRIBUTED ESTIMATOR DESIGN METHODS UNDER VARIOUS NOISE ASSUMPTIONS AND OPTIMIZATION CRITERIA

Method	Noise	Optimization
Distributed Kalman filtering	Random Noise	$\min \Theta_i$ s.t. $J_{e_i}^{BEC} \leq 0$
Distributed H_∞ filtering	Energy-Bounded Noise	$\min \gamma$ s.t. $J_{e,w,v}^\infty \leq 0$
Distributed set-membership filtering	UBB Noise	$\min \text{trace}(W_i)$ s.t. $J_{e_i}^{SM} \leq 0$

1) *Bounded Error Covariance*: For system (1) with multiplicative-type Gaussian white noise, the state estimation error covariance [36], [64], [77], [78] of estimator i is defined as

$$J_{e_i}^{BEC} \triangleq \mathbb{E}\{e_i e_i^T\} - \Theta_i$$

where Θ_i denotes a sequence of positive-definite matrices to be determined. Then, the main problem can be refined as designing suitable distributed estimators such that there exists a sequence of Θ_i satisfying $J_{e_i}^{BEC} \leq 0$. In this sense, the upper bound on the error variance of each estimator is guaranteed. Moreover, an optimality criterion lies in minimizing the bound Θ_i and eventually obtaining optimized estimators.

2) *H_∞ Performance*: For system (1) and (2) with additive-type energy-bounded noise w and v_i , the (average) H_∞ performance index [31], [32], [42], [43], [69] is defined as

$$J_{e,w,v}^\infty \triangleq \sum_{i=0}^{\infty} \left\{ \frac{1}{N} \sum_{i=1}^N \left\| Q_i^{\frac{1}{2}} e_i \right\|_2^2 - \gamma^2 \left\| R^{\frac{1}{2}} w \right\|_2^2 - \frac{\gamma^2}{N} \sum_{i=1}^N \left\| U_i^{\frac{1}{2}} v_i \right\|_2^2 \right\}$$

under zero initial conditions, where $\gamma > 0$ is a prescribed performance level, Q_i , R , and U_i are known sequences of positive-definite matrices. The response quality of the estimation error dynamics can be evaluated by ensuring $J_{e,w,v}^\infty \leq 0$ and a possible definition of optimality in this sense is to minimize the performance level γ .

3) *Set-Membership Performance*: For system (1) and (2) driven by UBB noise w and v_i , a common way to formalize a notion of performance and optimality is to develop a set of possible values enclosing the true system state and make this set the smallest set [79]. In this sense, the bounding performance index [23], [24], [61], [80]–[82] is defined as

$$J_{e_i}^{SM} \triangleq e_i^T(t+1)W_i^{-1}(t+1)e_i(t+1) - 1$$

at the next time instant $t+1$, where W_i is a sequence of positive-definite matrices to be determined. Then, under bounded initial condition and noise, by determining the ellipsoid $\mathcal{Z}_{t+1}^i := \{z(t) : J_{e_i}^{SM} \leq 0\}$, the actual estimate of the system's state on each sensor at each time step is guaranteed to be a bounding ellipsoidal estimation set containing all possible values of the true system state.

Depending on the aforementioned noise assumptions and performance criteria, several typical filtering methods that have been proposed for estimator design in the existing literature are shown in Table II.

III. DISTRIBUTED EVENT-TRIGGERED ESTIMATION

A. Motivation of Event-Triggered Mechanisms

A wireless-sensor node is often a microelectronic device which can only be equipped with a limited power resource [5]. The bandwidth of the wireless communication network sometimes may also be restricted. For example, in underwater communication environments, wireless channels allow low bit rates. However, to achieve a common tracking or monitoring objective, a wireless-sensor node needs to perform three main tasks: 1) data sampling; 2) data processing; and 3) data transmission. From the perspective of resource conservation, there is no doubt that traditional sampling and transmitting paradigms in a consecutive or periodical manner will lead to the inefficient usage of limited resources such as sensor power and bandwidth. On the other hand, it is a waste of communication and computational resources to perform consecutive data samplings and transmissions when there is a little fluctuation between two adjacent sampled data or when the system's state approaches its equilibrium and no external disturbance/noise is acting on the system. It is thus for these reasons that researchers are currently focusing on the design of distributed resource-aware estimation algorithms for sensor networks, among which the distributed event-triggered estimation has attracted considerable attention.

The concept of an event-triggered mechanism can date back to 1959 when Ellis claimed that “The most suitable sampling is by transmission of only significant data, as the new value obtained when the data are changed by a given increment” [83]. The rationale behind such a mechanism is that sampling and transmission actions are only invoked if the values of the signal of interest are significantly changed and, thus, communication and computation resources may be occupied only intermittently. The event-triggered mechanisms have now been proven to be a resource-efficient solution for distributed and networked systems; see [84]–[92] on networked control systems, and [93]–[100] on distributed multiagent systems. Indeed, an event-triggered mechanism is closer to human intelligence in such a way that the frequency of sampling actions or data transmissions is scheduled in an intelligent manner. In this sense, one can properly design the “events” in order to regulate the network communication and computational resources only when “needed” or “necessary.”

B. Principle of Event-Triggered Mechanisms

Denote the event release times on sensor i by a monotonically increasing sequence of time instants $\{m_i^j\}_{j=0}^\infty$ with $m_i^j \in \mathbb{N}$ and $m_i^0 = 0$. Then, the design of an event-triggered mechanism can be described as recursively calculating the event release time instant by

$$m_{t+1}^i = \inf_{t \in \mathbb{N}} \{t > m_t^i | e_i^T \Phi_i e_i > Th_i\}, \quad i \in \mathcal{V} \quad (12)$$

where $e_i^T \Phi_i e_i > Th_i$ denotes the predefined event-triggering condition involving an update error e_i , a positive weighting matrix Φ_i , and a well-defined threshold Th_i which can be customized as a non-negative constant, a time-dependent function, a function dependent on sensor measurements/estimates and

so on. In (12), the time difference between two consecutive event release instants is denoted as the interevent time, that is, $M_i^i \triangleq m_{i+1}^i - m_i^i$.

At each time instant t , sensor i checks whether its current measurement y_i or estimate x_i satisfies the event-triggering condition. If yes, the event release instant m_{i+1}^i is determined by (12) and the current measurement or estimate is sampled and/or transmitted. Otherwise, the current measurement or estimate will not be sampled and/or transmitted and the last released data will be utilized instead. Hence, it can be easily seen that $\{m_i^i\}_{i=0}^\infty \subseteq \mathbb{R}^+$ in the continuous-time case or $\{m_i^i\}_{i=0}^\infty \subseteq \mathbb{N}$ in the discrete-time case, thus significantly reducing the number of samplings and/or transmissions.

Recalling some existing results on distributed event-triggered estimation, there are two major categories of event-triggered mechanisms.

C. Event-Triggered Sampling Mechanisms

The system to be monitored by sensor networks is usually in the continuous physical world and its state is inherently analog signals. Nevertheless, virtually all distributed estimation algorithms are implemented in terms of software on digitalized computers. Besides, many advanced communication networks only permit digital information dissemination. Therefore, analog signals need to be sampled before they are disseminated through a communication network. Unlike the conventional time-triggered sampling mechanisms wherein the next sampling instant occurs after the elapse of a fixed period of time, event-triggered sampling only generates the next sampling instant when the signal value changes with a specified amount, namely, an “event” occurs. A key feature of event-triggered sampling lies in that the sampling instants are identical to the transmission instants. This means that sampling and data transmission occur simultaneously for each sensor. In this sense, the motivation of employing event-triggered sampling in sensor networks is to intelligently regulate the sampling actions on each sensor node only when some specific events take place and, thus, has the potential to alleviate the excessive usage of sensor power and network bandwidth for persistent sampling and transmission.

Note that in order to render the event-triggered sampling mechanism effective and efficient in practice, three important issues need to be carefully addressed.

1) *Minimal Interevent Time*: The minimal interevent time must be strictly greater than zero; otherwise, the so-called Zeno behavior [101] will occur and drive the system to sample extremely fast. In the continuous-time case, this is equivalent to guaranteeing the existence of a positive lower bound for all $\{M_i^i\}_{i=0}^\infty$. Whereas, finding a positive minimum interevent time in this case is generally difficult, especially when noise is present and/or the threshold Th_i in (12) is heavily dependent on neighboring sensors’ measurements/estimates. In the discrete-time case, however, the lower bound for $\{M_i^i\}_{i=0}^\infty$ can be ensured as “1,” thus straightforwardly excluding the Zeno behavior.

2) *Instantaneous Measurements/Estimates*: In the continuous-time case, the update error e_i in (12) is usually

predefined as some conditions related closely to instantaneous sensor measurements $y_i(t)$ or estimates $x_i(t)$. For example, $e_i(t) = \tilde{y}_i(t) - y_i(t)$ and $e_i(t) = \tilde{x}_i(t) - x_i(t)$ refines the update error as the measurement error and the estimation error with $\tilde{y}_i(t) = y_i(m_i^i)$ and $\tilde{x}_i(t) = x_i(m_i^i)$ representing the last released sensor measurement and estimate, respectively. In this sense, some extra hardware is required to instantaneously monitor $y_i(t)$ or $x_i(t)$, which inevitably increases the cost of system design and maintenance [102]. To overcome this issue, an alternative solution is to implement a self-triggered sampling mechanism so that the next event time instant is predicted based on the last triggered data and knowledge of the system dynamics. For more details on self-triggered sampling, we refer to [103] and [104].

3) *High Release Frequency*: In (12), if the event-triggering condition relies on the neighboring sensors’ latest triggered data, it may result in a high release frequency of events. For example, if the threshold Th_i heavily depends on a large number of neighboring sensors’ measurements/estimates, sensor i may trigger its sampled data not only at its own sampling instant m_i^i but also at any one of its neighbors’ sampling instants m_j^j as long as the triggering condition is satisfied. In this sense, the time-event time may become excessively small, thus leading to a high release of events.

D. Event-Triggered Transmission Mechanisms

Unlike event-triggered sampling, the sampling actions and data transmissions using an event-triggered transmission mechanism are orchestrated in a separate way but both take place at discrete instants of time. One may employ such an event-triggered transmission mechanism to intelligently regulate each sensor’s data transmissions through a communication network and, thus, potentially contribute to a reduction of sensor power expenditure and network bandwidth occupancy. Generally, sampling can be carried out in either a periodic manner or an aperiodic manner or even a random manner, while transmission is only invoked when some specific events occur. For example, by implementing a periodic sampling paradigm, each sensor’s measurement is synchronously sampled at equidistant instants of time $\{th\}$, where $t \in \mathbb{N}$ and $h > 0$ is a constant sampling period, while the event-triggering condition is verified at $\{th\}$ to decide whether or not to transmit the newly sampled data, which leads to the event release instants on each sensor of the form $\{m_i^i h\}$. Obviously, $\{m_i^i h\} \subseteq \{th\}$. Such an event-triggered transmission mechanism is also referred to as a *periodic event-triggered mechanism* [86] or a *sampled data-based event-triggered mechanism* [105]. Thus, the requirements of ensuring a strictly positive minimal interevent time (e.g., in the periodic sampling case, the minimal interevent time can be guaranteed as at least one sampling period h) and monitoring instantaneously the sensor measurements/estimates can be circumvented. However, similar to event-triggered sampling, one still needs to carefully deal with the high release frequency issue when a suitable event-triggered transmission mechanism is designed. Besides, it should be also pointed out that such an event-triggered transmission mechanism inevitably inherits some of the drawbacks

of conventional sampling mechanisms and may not contribute to a reduction of resource consumption on sampling because it only regulates the frequency of intersensor transmissions. For further reading, we refer the readers to the recent surveys [105] on event-triggered control for networked systems, [104] and [106] on event-triggered consensus for multiagent systems, and the references therein to enable them to obtain the state of the art in event-triggered mechanisms.

E. Classification of Distributed Event-Triggered Estimation

The update error e_i and the threshold Th_i are two key ingredients that leverage the idea of the event-triggered mechanism (12). A variety of definitions of the update error and threshold is used to distinguish the event triggered mechanisms in the literature. In the sequel, a classification of existing event-triggering conditions based on different update errors and thresholds is presented and some of the latest results on distributed event-triggered estimation over sensor networks will be reviewed accordingly.

1) *Event-Triggered Mechanisms Based on Constant Threshold*: In this case, the event-triggering condition is denoted as $e_i^T \Phi_i e_i > \delta_i$, where $\delta_i \geq 0$ represents the constant threshold and

$$\begin{aligned} e_i &= \tilde{x}_i - x_i, \mathcal{I}_i = y_i - C_i x_i \text{ [23], [107], [108];} \\ e_i &= \tilde{y}_i - y_i \text{ [109], [110];} \\ e_i &= \tilde{x}_i - x_i \text{ [58], [77], [111].} \end{aligned}$$

Such an event-triggered mechanism is also known as a *send-on-delta mechanism*. For example, in [23], the event-triggered mechanisms based on the constant threshold (ETM-C) is applied to regulate how frequently each sensor's innovation $\mathcal{I}_i$ should be transmitted to its neighbors when dealing with the problem of distributed set-membership filtering for a discrete time-varying nonlinear system subject to the UBB process and measurement noise. In [110], the problem of distributed Kalman consensus filtering is tackled for a discrete time-varying system subject to the Gaussian process and measurement noise, where the ETM-C is employed to regulate data communication through both sensor-to-estimator channels and estimator-to-estimator channels. By letting e_i be the measurement error $\tilde{y}_i - y_i$ and estimation error $\tilde{x}_i - x_i$, two types of event-triggering conditions are presented and distributed Kalman consensus filters based on triggered sensor measurements and estimates are designed. In [58], the event-triggered distributed filtering problem for nonlinear continuous-time Itô-type stochastic system is addressed, where the ETM-C is adopted to determine when each sensor's local estimate should be broadcast to its neighbors. An adaptive algorithm to determine the triggering threshold is developed, which preserves the exponential mean square stability of the resulting filtering error system while achieving a desired level of the average transmission rate. In [77], the ETM-C is employed to tackle the problem of distributed Kalman consensus filtering for two-target tracking sensor networks, where the update error e_i is refined as estimation errors on sensors for two targets.

2) *Event-Triggered Mechanisms Based on Instantaneous Measurement-/Estimate-Dependent Threshold*: In this case, the event-triggering condition is rewritten as $e_i^T \Phi_i e_i >$

$\sigma_i f_i^T \Psi_i f_i$, where $\sigma_i \geq 0$ denotes the threshold parameter, and f_i represents the instantaneous measurement-/estimate-dependent term and

$$\begin{aligned} e_i &= \tilde{x}_i - x_i, f_i = x_i, \Psi_i \equiv \Phi_i \text{ [112];} \\ e_i &= \tilde{x}_i - x_i, f_i = x_i, \Psi_i \equiv \Phi_i \equiv I \text{ [59];} \\ e_i &= \tilde{\mathcal{I}}_i - \mathcal{I}_i, \mathcal{I}_i = y_i - C_i x_i, f_i = \mathcal{I}_i, \Psi_i \equiv \Phi_i \text{ [113];} \\ e_i &= \tilde{\mathcal{I}}_i - \mathcal{I}_i, \mathcal{I}_i = y_i - C_i x_i, f_i = \mathcal{I}_i, \Psi_i \equiv \Phi_i \equiv I \text{ [60].} \end{aligned}$$

For example, in [112], the problem of distributed H_∞ consensus filtering is studied for a discrete time-varying stochastic system over a finite horizon, where the event-triggered mechanisms based on instantaneous measurement-/estimate-dependent threshold (ETM-I) is adopted for each sensor node to reduce the communication burden of local estimates. The problem of event-triggered distributed H_∞ filtering is addressed in [60] for a discrete time-delayed system in the presence of packet dropouts and stochastic nonlinearities. The ETM-I is employed to alleviate the extravagant resource consumption on transmitting sensors' innovations.

3) *Event-Triggered Mechanisms Based on Released Measurement-/Estimate-Dependent Threshold*: In this case, the event-triggering condition is described by $e_i^T \Phi_i e_i > \sigma_i f_i^T \Psi_i f_i$, where f_i now denotes the sampled and released measurement-/estimate-dependent term and

$$\begin{aligned} e_i &= \tilde{x}_i - x_i, f_i = \sum_{j \in \mathcal{N}_i} a_{ij}(\tilde{x}_i - \tilde{x}_j), \Psi_i \equiv \Phi_i \text{ [31];} \\ e_i &= \tilde{y}_i - y_i, f_i = \tilde{y}_i, \Psi_i \equiv I \text{ [61], [81];} \\ e_i &= \tilde{\mathcal{I}}_i - \mathcal{I}_i, \mathcal{I}_i = \sum_{j \in \mathcal{N}_i} a_{ij} y_j, f_i = \tilde{\mathcal{I}}_i, \Psi_i \equiv \Phi_i \text{ [22];} \\ e_i &= \tilde{\mathcal{I}}_i - \mathcal{I}_i, \mathcal{I}_i = y_i - C_i x_i, f_i = \tilde{\mathcal{I}}_i, \Psi_i \equiv \Phi_i \equiv I \text{ [114].} \end{aligned}$$

For example, in [31], the event-triggered mechanisms based on the released measurement-/estimate-dependent threshold (ETM-R) are applied to solve the distributed H_∞ consensus filtering problem for a discrete linear time-invariant system. The threshold parameter σ_i is time varying with attainable upper and lower bounds, and the weighting matrix Φ_i is assumed to be threshold-parameter-dependent. Then, a threshold-parameter-dependent approach is developed to establish criteria on the existence of desired consensus filters while ensuring the asymptotic stability of the resulting filtering error system under the prescribed H_∞ performance. In [81], by resorting to the above ETM-R, a distributed set-membership filtering method is proposed to detect the islanding fault for distributed grid-connected solar photovoltaic (PV) generation systems. In [22], the problem of distributed event-triggered H_∞ filtering for a continuous linear time-invariant system with communication delays is studied. Each sensor can decide when its fused measurement should be transmitted to its neighbors. A co-design algorithm is then presented to carry out the trade-off analysis between communication resource utilization and desirable H_∞ performance.

It is noteworthy that the event-triggered mechanisms based on the triggering conditions specified above are commonly known as *static event-triggered mechanisms*, that is, the threshold either is fixed as δ_i or follows a static expression $\sigma_i f_i^T \Psi_i f_i$ on the sensor measurements/estimates. In this sense, an event is triggered once $e_i^T \Phi_i e_i$ surpasses the static threshold δ_i or $\sigma_i f_i^T \Psi_i f_i$, which essentially represents a conservative solution generating unnecessary samplings and/or data transmissions. For example, one may introduce a time-dependent

function $\rho_i e^{-\varrho_i t}$ (where $\rho_i \geq 0$, $\delta_i + \rho_i > 0$ and $\varrho_i > 0$) into the static threshold δ_i or simply insert a small positive constant into the static threshold $\sigma_{if_i}^T \Psi_{if_i}$ such that $e_i^T \Phi_i e_i$ does not frequently exceed the static threshold, which definitely further decreases the total number of released events. A thorough review of these event-triggered mechanisms is provided in the recent surveys [104], [106] in the context of multiagent coordination control. However, it should be emphasized that when the threshold involves an additional positive constant, the stability of the resulting estimation error system needs to be re-evaluated in the boundedness sense.

4) *Dynamic Event-Triggered Mechanisms*: A more general way to reduce the conservatism introduced by static event-triggered mechanisms is to dynamically adjust the threshold by inserting an offset variable (or called an internal dynamic variable [115]), which leads to the dynamic event-triggered mechanism (DETM). In this case, the event-triggering condition is revised as $e_i^T \Phi_i e_i > \sigma_{if_i}^T \Psi_{if_i} + (\lambda_i / \theta_i)$. Here, λ_i is referred to as an auxiliary offset variable satisfying the following dynamics $\lambda_i(t+1) = \rho_i \lambda_i(t) - e_i^T \Phi_i e_i + \sigma_{if_i}^T \Psi_{if_i}$ [24], where $\lambda_i(0) \geq 0$ is the initial condition, $\rho_i \in (0, 1)$ and $\theta_i \geq 1/\rho_i$. In [24], such a DETM with $e_i = \tilde{y}_i - y_i$, $f_i = \tilde{y}_i$, and $\Psi_i \equiv \Phi_i$ is implemented to cope with the distributed set-membership filtering problem for a discrete linear time-varying system under the influence of the UBB process and measurement noise. A recursive convex optimization algorithm is proposed to determine the optimal bounding ellipsoidal estimation set on each sensor which encloses the system's true state. Moreover, it is theoretically and numerically shown that the DETM can produce larger average interevent time and, thus, less totally released data packets than the static mechanism.

IV. APPLICATIONS OF DISTRIBUTED EVENT-TRIGGERED ESTIMATION

In this section, we briefly outline two applications of distributed event-triggered estimation methods which have received great attention in practical sensor network-based monitoring systems including distributed grid-connected generation systems and target tracking systems.

A. Distributed Grid-Connected Solar PV Generation Systems

Unintentional islanding is regarded as one of the most significant safety issues in distributed grid-connected generation systems, including wind and solar PV generation systems. Unintentional islanding operation is an undesirable mode of operation where the power supply from the main grid is interrupted for various reasons, such as line tripping, equipment fault/failure, circuit-breaker (CB) misoperation, human errors, and external malicious attacks, but the generation system continues to supply power to the end users or local loads. Islanding can result in electrical damage to end-user equipments, poor power quality, and even safety hazard to the utility service personnel, which thus must be identified as soon as possible. There are several islanding detection methods in [116]–[121]. In [81], an application of a distributed

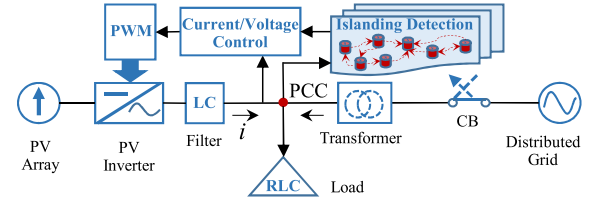


Fig. 3. Grid-connected PV generation system configuration at normal operation mode (CB is closed) and islanding operation mode (CB is open).

event-triggered set-membership estimation method on islanding detection in distributed grid-connected solar PV generation systems, as shown in Fig. 3, is investigated.

The distributed event-triggered set-membership estimation problem can be cast into designing a suitable event-triggered mechanism as well as distributed event-based estimators such that an ellipsoidal state estimation set $\mathcal{S}_{t+1}^i$ can be determined at every instant of time which always encloses the true state of the system under any values of UBB disturbance input, measurement noise, and initial condition, where

$$\mathcal{S}_{t+1}^i \triangleq \left\{ s(t+1) : e_{t+1}^{iT} (P_{t+1}^i)^{-1} e_{t+1}^i \leq 1 \right\} \quad (13)$$

with $e_{t+1}^i = s(t+1) - x_i(t+1)$ and P_{t+1}^i representing a real-valued positive time-varying matrix sequence.

The core of the islanding detection scheme lies in recursively calculating the two ellipsoidal sets: 1) the state estimation set $\tilde{\mathcal{S}}_{t+1}^i$ without islanding in the PV generation system and 2) the actual state estimation set $\mathcal{S}_{t+1}^i$ to be detected. The islanding detection scheme can then be formulated as checking the existence of the intersection between the two sets under the following decision law:

$$\begin{cases} \tilde{\mathcal{S}}_{t+1}^i \cap \mathcal{S}_{t+1}^i \neq \emptyset & \Rightarrow \text{no alarm} \\ \tilde{\mathcal{S}}_{t+1}^i \cap \mathcal{S}_{t+1}^i = \emptyset & \Rightarrow \text{alarm of islanding.} \end{cases}$$

The principle behind such a detection scheme is that the occurrence of islanding may result in a deflected center of the actual state estimation set $\mathcal{S}_{t+1}^i$. In this case, if the intersection is empty, islanding occurs, which, however, is not always true otherwise. This is because the islanding may hide in the disturbance/noise or the effect of islanding may not cause sufficient deviation of the centers of the two sets, thus the two sets may still contain intersections. From this perspective, the detection scheme above may represent a conservative solution and necessitates further improvement. On the other hand, the comparative studies between time-triggered mechanisms and event-triggered mechanisms in [81] show that there is a tradeoff of islanding detection performance and communication performance. More specifically, the islanding detection performance of a time-triggered mechanism is much better than that of an event-triggered mechanism while the event-triggered mechanism contributes to the reduction of superfluous data transmissions and, thus, unnecessary communication resource usage.

B. Target Tracking Systems

Successful target tracking depends on several factors but the key one is how accurately it can extract useful target state

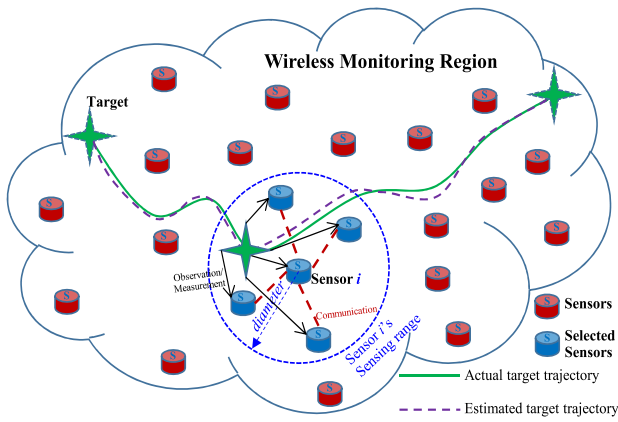


Fig. 4. Mobile target tracking using a sensor network.

information (e.g., position and velocity) from local observations recorded by spatially dispersed sensors. On the one hand, this key factor is related to the choice of an appropriate mathematical model to represent the target motion since a good target model will certainly facilitate this information extraction to a great extent [122]. For various mathematical models of target motion/dynamics and interrelationships among the models, interested readers are referred to the survey paper [122]. On the other hand, it essentially lies in the design of an effective tracking algorithm to estimate the target's state as accurate as possible because sensor observation data are usually restricted. Fig. 4 demonstrates a wireless-sensor network-based target tracking system consisting of a mobile target and a number of wireless sensors. Due to limited power, computational capacities, and memory, sensors can only communicate intermittently their local observations and/or estimates with their neighboring sensors residing in their sensing ranges. Moreover, each sensor cannot only observe the moving target but also gather local observations and/or estimates from its neighbors to form an updated estimate so that it can predict the future state of the target. For this reason, a key issue is to develop a suitable event-triggered mechanism which can efficiently schedule intersensor communication in order to reduce unnecessary interactions among the neighboring sensors. For example, in [58], the positioning problem of a wheeled mobile robot is investigated via distributed event-triggered filtering over a wireless-sensor network. An adaptive threshold algorithm is proposed to achieve a desired average transmission rate during the estimation and tracking processes. In [61], a distributed event-triggered set-membership filtering approach, combined with a partial information transmission scheme, is presented for tracking a dynamic target whose position and velocity evolve according to a discrete-time linear time-invariant model. It is shown that more measurement transmissions can result in better estimation and tracking performance. In [77], an event-triggered Kalman-consensus filtering approach is studied for two target tracking sensor networks. Two groups of sensors can both track their corresponding targets while maintaining satisfactory communication frequencies.

Depending on practical applications, sensors can be deployed in a predetermined or mobile manner, which

leverages an idea of static sensor networks or mobile sensor networks. Apparently, mobile sensor networks are much more versatile than static sensor networks as sensor nodes can be deployed in any scenario and accommodate rapid topology changes [123]. Meanwhile, mobile sensor network protocols and algorithms must possess self-organizing capabilities [5]. In the context of event-triggered target tracking, this necessitates an intelligent event-triggered mechanism which can identify the time-varying sensor interaction topology. To the best of the authors' knowledge, the target tracking problem using distributed event-triggered estimation that is dependent on a time-varying sensing/communication topology has not been adequately addressed.

V. CONCLUSION

Recent results of distributed event-triggered estimation for dynamical systems over sensor networks have been systematically reviewed and examined. Two application examples of distributed event-triggered estimation have also been outlined. In what follows, some important and yet challenging issues worthy of further investigations are suggested.

1) *Scalability and density*: Depending on the applications, the number of actual sensor nodes deployed in a monitoring area may be of the order of hundreds or even higher. Most of the existing distributed event-triggered estimation algorithms have the scalability issue that the computational complexity of the algorithms dramatically increases when the number of nodes goes larger. Another issue closely related to scalability is the density of the sensor networks. For example, the density can range from a few sensors to hundreds of sensors in a region less than 10 meters in diameter [5]. Therefore, novel distributed event-triggered estimation algorithms must be able to effectively and efficiently tackle the scalability and density issues.

2) *Guaranteed convergence rate in fixed/prescribed-time* [124]–[126]: The convergence rate is regarded as one of the most important performance indices of estimation and control algorithms. There is no doubt that fast convergence should be ensured in practice when distributed estimation solutions are applied. For example, fast convergence is a prerequisite to meet the rapid response requirement of power grids. However, the distributed fixed-/prescribed-time estimation problem under event-triggered sampling/transmission has not been well addressed in the literature. How to carefully design an event-triggered mechanism over the fixed/prescribed interval such that the fixed-/prescribed-time estimation performance can be achieved while maintaining satisfactory resource occupancy deserves a deep investigation.

3) *Energy-efficient node selection/activation/scheduling*: Although an event-triggered mechanism enables resource-awareness sampling and data transmission solutions for sensor network-based monitoring systems, the energy reduction on samplings and transmissions is insufficient in sensor networks because the main cause for energy consumption of a wireless-sensor node lies in its idle listening for a message [106], [127]. How to further develop energy-efficient distributed estimation techniques in such a way

to intelligently select/activate/schedule tasking sensors while minimizing energy consumption and preserving desired estimation performance requires further research effort.

4) *Distributed secure estimation and attack detection*: The wireless communication medium increases the flexibility and scalability of the entire sensor network; meanwhile, bringing more opportunities to security threats, which renders distributed secure estimation fundamentally significant in sensor networks [128]–[131]. However, some existing results [62], [63], [132] assume no difference between random data losses (or missing measurements) and denial-of-service attacks (or jamming attacks) when establishing attack models to quantitatively analyze the performance degradation. Therefore, how to expose the active behaviors of attackers and the passive impacts of data losses should be further explored. Moreover, to control the damage caused by attacks to the overall system within a tolerant limit, it is essentially desirable to develop suitable detect mechanisms that can locate and identify the attacks in a timely fashion [133], and further warn the defender to allow corrective actions. Arguably, the following two attack detection approaches have been widely employed in the context of a distributed secure estimation framework: 1) χ^2 -detector approach based on statistical analysis [134]–[136] and 2) *fault detection and isolation approach* based on residual analysis [62], [137]. However, such a χ^2 -detector approach requires that the attack signals obey Gaussian distribution [82] and may fail to detect sophisticated deception attacks with good knowledge of the system. How to devise effective detection approaches for sophisticated attacks such that their impacts on the system performance can be minimized remains a challenging issue. The widely used attack detection approach based on fault detection, on the other hand, treats attacks and faults in the same manner, which greatly simplifies the estimation and detection procedures. Nevertheless, from the defender's perspective, one needs to distinguish faults from malicious attacks such that a proper corrective action can be initiated. Strictly speaking, it is not always possible to differentiate between faults and attacks because a cunning adversary can intentionally launch an attack that behaves like a fault. Besides, the correlated effects of attacks and some uncertain factors, such as the unknown process and measurement noise and random network communication environment, on sensor measurements make it potential to falsely detect and alarm an attack. Hence, how to establish a rigorous theoretical framework of distributed estimation and attack detection in order to effectively differentiate the effects caused by malicious attacks from faults is a promising research topic. Apart from the secure estimation and detection strategies, the corresponding compensation measures should also be addressed from the perspective of control [138], [139].

5) *Privacy-preserving distributed estimation*: Most existing distributed estimation strategies require sensors to continuously release private (local) observations to the others in the vicinity. This can result in an undesirable loss of privacy for participating sensors. For example, in an intelligent traffic monitoring system, individual position traces are identified by smart phones while personal information, such as a person's residence or work location can be revealed as well. Thus,

distributed privacy-preserving estimation in a sensor network setting represents an interesting research topic. The tradeoff between sensor privacy and system performance is also a significant issue.

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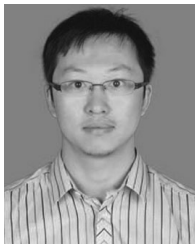
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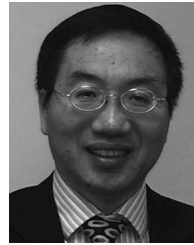
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